

ADITYA RAI

UNREAL ENGINE DEVELOPER

📞 6387725044 ✉ brodyrai16@gmail.com [linkedin.com/in/aditya-rai10](https://www.linkedin.com/in/aditya-rai10) github.com/Adityajr10 Bangalore

SUMMARY

A highly passionate Unreal Engine/Game developer bringing a proven portfolio of various projects and games. With over 3 years in game development, I specialize in core game logic, combat & enemy AI systems, gameplay architecture, and implementing complete game flow in C++ and Blueprints.

PROJECTS

TheLastBlood — Souls-like Combat AI System (Unreal Engine, C++) ▶ [Demo](#)

Enemy AI & Boss Systems (Elden Ring-inspired):

- Built a layered AI "brain" that reads the fight in real time — 3-stage decision-making (stance → action → ability) with personality profiles, weighted scoring, and priority tiers, plus a pattern tracker that learns the player's dodge/parry habits to bait and punish them.
- Bosses run a designer-authored Attack Director with reactive combo edges modeled on Elden Ring's follow-up logic, and a rule-based Skill System replicating its boss weapon-arts — powering multi-phase boss kits (6+ unique skills each), pack AI coordination, and adaptive weighting so bosses abandon moves the player has solved.

Player Systems:

- Engineered a Souls-like combat framework where every input flows through one authoritative 14-state machine — buffered combo chains, i-frame dodges, and rage/stamina economies.
- Recreated Elden Ring's defense feel with impact-timed parry/block resolution and posture meters that turn guard-breaking into a win condition, plus a synced animation system with motion warping for cinematic finishers, executions, and grabs.
- Delivered a data-driven progression architecture — per-weapon skill trees gating a 15+ ability roster, 8-slot loadouts, save persistence — where shipping a new weapon requires zero code changes.

Dragon AI — Personality-Driven Combat System (Unreal Engine, C++) ▶ [Demo](#)

- Built a dragon AI whose combat decisions emerge from six personality traits, not scripted logic — traits drive a 7-instinct layer that filters abilities before any move is chosen.
- Implemented a real-time weighted-scoring system (distance, energy, cooldowns, altitude, courage-vs-health) that picks the final move each decision cycle.
- Abilities run on a modular, data-driven architecture, so new moves are added via data, not code — every dragon feels genuinely unique, powered by one scalable system.

Arka (PS5 Platform — Company Project) (Unreal Engine, C++)

- Architected an interface-driven Interaction system (IInteractionInteractable) decoupling GlobalInteractionComponent from all actor types, with real-time IK hand-alignment using Two Bone IK, FABRIK, and CCDIK.
- Built a delegate-driven Inventory system where InventoryComponent never references UI directly — DataTable acts as single source of truth for item names, icons, thresholds, and attribute upgrades with zero hardcoded C++.
- Engineered a Puzzle system with zero per-frame Tick overhead — timer-based state updates fire only on interaction and ring axis movement driven through math, not animation.
- Designed enemy AI architecture using Behaviour Trees (engage, attack, strafe) and an InPlaceRotation system (RDR2-style) built as a reusable standard for both creature and humanoid agents.
- Developed Player systems using GAS for abilities and combat, alongside custom Health, Swimming, Zipline, and Resident Evil-style 3D Inspection systems — each as a self-contained UActorComponent.
- Resolved a 6-day team-wide build failure through root-cause analysis and owned CI/CD pipeline; authored component-based, delegate-driven architecture standards adopted across a 30-person team.

Golf Simulator (AUTOMATION) ▶ [Demo](#)

- Developed an automation system to procedurally generate more than 40,000 golf courses with a single click, enabling seamless creation and saving of courses.

INNOVATIVE PROJECTS

VR/AR Projects ▶

VR Cave Bat Shooter.

Zombie Shooter ▶

TPS, Buddy AI, Zombie AI, Cinematics

Vehicle Construction Automation ▶

AR REPLICA ▶

Developed a playable AR character with adjustable height, speed, and rotation, featuring interactive object logic in Unreal Engine.

CORE STRENGTHS

- COMBAT & ENEMY AI SYSTEMS
- GAMEPLAY ABILITY SYSTEM (GAS)
- ANIMATION & IK (FABRIK, MOTION WARPING)
- DATA-DRIVEN ARCHITECTURE

AWS / DEVOPS

- Kubernetes/Docker microservices orchestration – CI/CD with Jenkins, GitLab CI & ArgoCD.
- AWS: EC2, S3, IAM, RDS, Lambda.
- Monitoring: Prometheus, Grafana, ELK.
- Configuration: Ansible, Puppet, Chef.
- IaC: Terraform & CloudFormation

SKILLS

- UNREAL ENGINE
- C++ AND BLUEPRINT
- AUTOMATION
- GAMEPLAY SYSTEMS
- PLUGIN DEVELOPMENT
- PROCEDURAL GENERATION
- PERFORMANCE OPTIMIZATION

LANGUAGES

- C++
- C#

EDUCATION

Mtech (Integrated) (CSE)

2019–2024 (NIET)

GRAPHICS / ENGINE SKILLS ▶

- Perforce, Git, GitOps
- Vulkan
- SFML, Raylib, SDL
- DirectX

- Contains over 50,000 lines of C++ code to handle the creation of course elements like bunkers, fairways, greens, creeks, and pathways.
- Built plugins, including a Dynamic Spawning System and Height Adjuster, for enhanced automation and customization.
- Transformed raw JSON data into Unreal Engine coordinates to generate splines and shapes for course components.

Cricket VR (Meta Link) ▶ Demo

- Built "TOK Premier League" VR game with precise bat physics and full shot range (blocks, drives, cuts, pulls, sweeps).
- Added dynamic footwork, haptic feedback, edge detection, and physics-based bowling with swing.
- Featured AI fielders for immersive cricket gameplay on Meta Quest.

GYM SIMULATOR ▶ Demo

- Created a gym simulation game; developed features for gym ownership, including allowing or restricting entry, cleaning and placing gym equipment.
- Implemented AI-driven gym-goer behavior with characters performing different exercises, resting, recharging, and transitioning between activities dynamically.
- Designed a realistic gym environment with interactive elements for an engaging player experience.

UNREAL PLUGIN DEVELOPMENT (Tools and Optimization) ▶ Demo

- Developed multiple plugins for Unreal Engine, including a Plantation System, Advanced Blueprint Helper, and Optimized Landscape System.

THE WOLF ▶ Demo

- A survival game centered around a wolf's fight for food and survival.
- Implemented AI-driven animal behaviors where animals can form alliances, attack, or coexist based on interactions.
- Integrated features like reproduction, aging, and growth, with dynamic group hunting and leadership mechanics.

FIRST PERSON OBSTACLE SHOOTING ▶ Demo

- Created a shooting game with dynamic, progressively challenging obstacles using C++ and Blueprints. Designed levels with varying obstacle movements, including zigzag and random axis shifts.

EXPERIENCE

Rickshaw Studio **June-25 Till Present**

Game Systems Programmer (Unreal Engine, C++)

- Built a real-time puzzle interaction system using FABRIK and Two-Bone IK for hands-on object manipulation.
- Designed the enemy AI architecture defining core behavior and decision-making for all game enemies.
- Developed an in-place rotation system (RDR2-style) enabling natural turn-in-place for any creature.
- Created the player ability system using GAS — magic abilities and combat framework.
- Built interaction, inspection & inventory systems — Resident Evil-style item inspection, with a single interaction component handling real-time interaction with any object.

Teams of Keys **March-24 To May-25**

Game Developer

- Writing C++ classes for game creation and automation.
- Integrating Blueprint with C++ for seamless gameplay features.
- Developing games that involve game design, game flow, and client projects.
- Plugin creation; core mechanisms fully based on C++.

Zapnosys AI Pvt Limited **December-23 To March-24**

Unreal Engine Developer

- Developing Unreal video engine using C++.
- All functions, emotions and characteristics are created in C++.
- Implementing our own Interpreter.

Freelance (Unreal Engine) **April-23 Till October-23**

- Development of games and gameplay mechanisms using Unreal Engine.